

William J. Scott

Software Engineer • Sacramento, CA • (360) 303-1563
wsc0tt@outlook.com • github.com/wsc0tt • linkedin.com/in/wscottdev

Professional Summary

Software engineer with hands-on experience spanning systems programming, C/C++, and full-stack development. Comfortable working in Linux/Unix environments with a solid foundation in networking fundamentals and software testing practices. Leadership experience managing complex projects and delivering results under real-world constraints.

Education

California State University, Sacramento, CA — B.S. Computer Science, December 2025
GPA: 3.8 • Dean's Honor List

Relevant Coursework

Systems Programming (Linux/Unix), Operating Systems, Computer Architecture, Computer Networking, Parallel Programming (CUDA), Object-Oriented Programming, Data Structures & Algorithms, Intelligent Systems (AI/ML)

Technical Skills

Languages: C, C++, Python, Java, SQL, TypeScript, JavaScript

Systems & Tools: Linux/Ubuntu, pthreads, Git, Docker, Make, CUDA

AI/ML: TensorFlow, Transfer Learning, Model Optimization, Claude

Backend: PostgreSQL, RESTful APIs, Supabase, .NET

Cloud & DevOps: Github Actions, AWS, Azure

Engineering Projects

Multithreaded Memory Management Hasher | C, pthreads, OS Concepts

- Built a thread-safe memory allocator in C achieving 250x speedup over a serialized baseline through fine-grained locking and per-thread memory pools.
- Optimized contention by migrating from coarse-grained global locks to lock-free fast paths, reducing latency in shared-resource access.

Mars Rover Terrain Risk Classifier | Python, TensorFlow

- Engineered a lightweight CNN (EfficientNetB0) for terrain risk classification targeting deployment on resource-constrained onboard rover hardware.
- Achieved 82.6% classification accuracy with an 87.6% reduction in model size relative to baseline, optimizing for tight memory and compute budgets typical of embedded systems.
- Designed a rule-based heuristic converting pixel-level segmentation into traversability scores, enabling real-time risk assessment within strict latency targets.

Maritime Engineers Union – Job Management Application | React, PostgreSQL, Supabase, Docker, Jira

- Co-led a 9-engineer team through a full software development lifecycle, delivering a production application for the Maritime Engineers Beneficial Association (MEBA), a labor union serving 20,000+ members.
- Designed the PostgreSQL schema including tables, relationships, and constraints, and authored SQL queries, Postgres functions, and database triggers to support production operations.
- Built Supabase Edge Functions for serverless backend logic and configured authentication flows, REST API integrations, and real-time subscriptions to push live updates to connected clients.

Professional Experience

Syblon Reid — Laborer Foreman | Folsom, CA July 2013 - November 2022

- Led 30+ infrastructure projects end-to-end for high-profile utility and government clients including PG&E, SMUD, DWR, and Southern California Edison, with zero safety incidents.
- Solved critical logistics bottlenecks independently under 48-hour deadlines on active utility infrastructure.
- Adapted workflows daily to shifting site conditions, exercising rapid decision-making under incomplete information.